MAHDI MOEINI

mmoeini@deloitte.ca | (709)-699-9147 | Ottawa, Canada | linkedin.com/in/mmoeini | m-moeini.github.io

WORK EXPERIENCE

Consultant - Machine Learning Engineer, *Deloitte* **Mar 2025 – Present**

A consulting firm providing AI solutions, along with audit, tax, and risk advisory services to clients across industries.

- Collaborated with cross-functional teams—including data engineers, analysts, and business stakeholders—to translate complex requirements into deployable machine learning (ML) solutions.
- Developed and maintained production-quality code using programming and scripting languages such as Python, PySpark, and SQL, ensuring efficient implementations for data pipelines and machine learning and deep learning models.
- Extracted data from structured and unstructured sources, preprocessed it using appropriate data preparation techniques.
- Trained and refined various ML models, and applied them to solve targeted business problems.
- Deployed cloud-hosted ML pipelines using MLOps best practices to ensure scalable, and production-ready applications.
- Monitored machine learning models through structured evaluation pipelines and visualized performance metrics in a business-friendly manner to help stakeholders quickly understand the model behavior and make informed decisions.

SKILLS: *Python, PySpark, SQL, Azure, Databricks, MLflow, Delta Lake, Docker, Scikit-learn, PyTorch*

AI/ML Researcher, *Memorial University of Newfoundland, St. John's, Canada* **Sep 2022 – Sep 2024**

- Explored the possibility of controlling a wheelchair through "singing imagination" by designing a Brain-Computer Interface (BCI) experiment to collect Electroencephalography (EEG) time-series datasets from 15 participants
- Achieved 85% accuracy in classifying EEG signals using various deep learning and machine learning models (CNNs, RNNs, SVM, XGBoost) and provided insights into model performance through statistical analysis
- Enhanced data quality and the labeling system by building a data preprocessing pipeline using MATLAB & Python
- Boosted classification accuracy by 8% using efficient feature extraction & optimization methods (CSP, mRMR)
- Improved data analysis speed by 50% via building an automated pipeline on Compute Canada Cloud (Code)

SKILLS: *Python, R, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Matplotlib, Seaborn, EDA, AWS*

Software Developer, *Toranj Innovation Technologies, Tehran, Iran* **May 2021 – Jun 2022**

A health tech company concentrated on creating software solutions for the healthcare industry.

- Increased the customer base by 20 % by expanding system scalability through developing fault-tolerant Java microservices
- Doubled the speed of User Acceptance Testing process by implementing RESTful APIs integrated with FitNesse server
- Eliminated the need for direct SQL queries by utilizing Object-Relational Mapping (ORM) systems like JPA
- Contributed to software development in a CI/CD environment by utilizing Git for version control

SKILLS: *Java, Spring Boot, JPA, SQL, PostgreSQL, ETL, Docker, Git, Microservice Architecture, MVC Pattern*

EDUCATION

- **Master's Degree - Computer Engineering,** *Memorial University of Newfoundland* **Sep 2022 – Sep 2024**
- **Bachelor's Degree - Electrical Engineering,** *Iran University of Science and Technology* **Sep 2017 – May 2022**

CERTIFICATIONS & SKILLS

- **Certifications:** AWS Certified Cloud Practitioner, Coursera Machine Learning Specialization
- **Programming Languages**: Python, Java, C, C++, MATLAB, SQL, R
- **Industry Skills**: Machine Learning, Deep Learning, MLOps, Big Data (Spark), Data Science, Data Visualization, Statistics, Object-Oriented Programming (OOP), SOLID Programming, AWS, Azure, Databricks